CHEMISTRY

SECTION – I

Single Correct Choice Type

This section contains 4 single correct choice type questions. Each question has 4 choices: (a), (b), (c) and (d) for its answer, out of which **ONLY ONE** is correct.

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1. A sample of sea water containing 2.8% NaCl by mass has density 1.03 g/cm^3. A saturated solution of NaCl in water is 5.45 M. How much water would have to be evaporated from 4 L of sea water before NaCl would start precipitating?

(a) 3.64 L (b) 0.36 L (c) 2.64 L (d) 1.36 L

2. A silver electrode is immersed in a saturated $Ag_2SO_4(aq)$ solution. The potential difference between silver electrode and standard hydrogen electrode is found to be 0.74 V. The K_{sp} of Ag_2SO_4 would be (Given that $E^{o}_{Ag^+/Ag} = 0.799$ V)

(a) 5×10^{-6} (b) 5×10^{-5} (c) 5×10^{-4} (d) 5×10^{-3}

3. $CH_3-\overset{\overset{O}{\|}}{C}-Cl + CH_3-CH_2-\underset{\underset{CF_3}{|}}{CH}-COOEt \xrightarrow[(ii)\ H^+/H_2O]{(i)\ OH^-} (A) \xrightarrow{\Delta} (B) \xrightarrow{C_6H_5-\overset{\overset{O}{\|}}{C}-\overset{18}{O}-O-H} (C)$

Product (C) would be

(a) $CH_3-\overset{\overset{O^{18}}{\|}}{C}-O-\underset{\underset{CF_3}{|}}{CH}-C_2H_5$ (b) $CH_3-O-\overset{\overset{O^{18}}{\|}}{C}-\underset{\underset{CF_3}{|}}{CH}-C_2H_5$

(c) $CH_3-\overset{\overset{O}{\|}}{C}-\overset{18}{O}-\underset{\underset{CF_3}{|}}{CH}-C_2H_5$ (d) $CH_3-\overset{18}{O}-\overset{\overset{O}{\|}}{C}-\underset{\underset{CF_3}{|}}{CH}-C_2H_5$

4. Naturally occuring potassium contains 1.2×10^{-2} % of ^{40}K. The human body contains 3.5×10^{-1} % of potassium by weight. The total radioactivity resulting from ^{40}K decay in 80 kg human would be ($t_{1/2}$ for $^{40}K = 6.9 \times 10^{14}$ min^{-1})

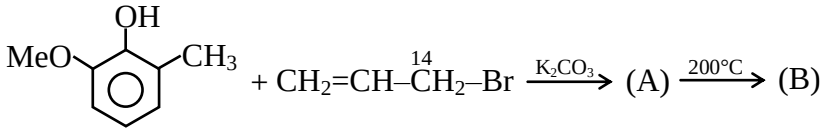
(a) 5.06×10^3 dpm (b) 5.06×10^4 dpm
(c) 5.06×10^5 dpm (d) 5.06×10^6 dpm

SECTION – II
Multiple Correct Choice Type

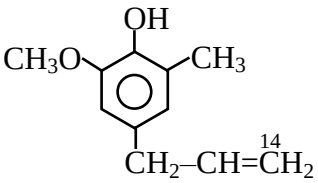
This section contains 4 multiple correct choice type questions. Each question has 4 choices: (a), (b), (c) and (d) for its answer, out of which **ONE OR MORE** is/ are correct.

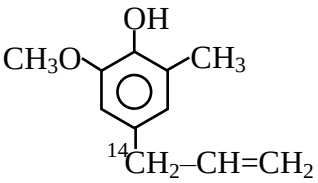
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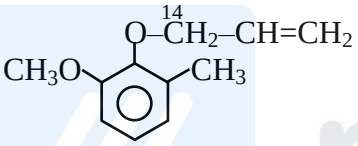
5. Product (B) of the following reaction

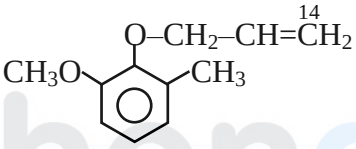


would be

(a) 

(b) 

(c) 

(d) 

6. An inorganic compound (A) is formed by passing a gas (B) through a concentrated liquor containing sodium sulphide and sodium sulphite. On adding (A) into dilute solution of $AgNO_3$, a white precipitate appears which quickly changes into black coloured compound (C). Which of the following is/are correct statement(s)?

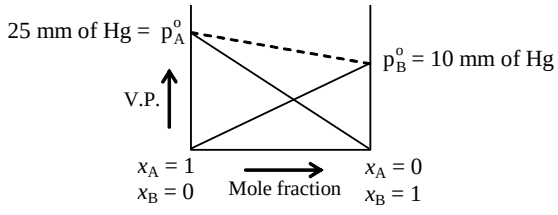
(a) (B) is NH_3 and (C) is $(NH_4)_2S$.

(b) (A) is $Na_2S_2O_3$ and white precipitate is of $Ag_2S_2O_3$.

(c) (B) is I_2 and (C) is Ag_2S.

(d) (A) is Na_2SO_4 and white precipitate is of Ag_2SO_4.

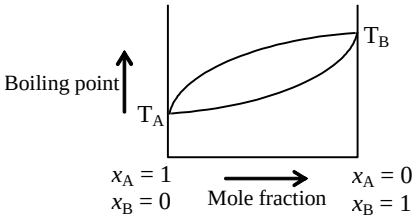
7. Two liquid (A) and (B) when mixed in any proportion forms an ideal solution. A graph vapour pressure vs mole fraction for this solution is shown as under:



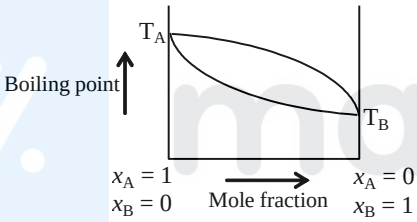
Which of the following statement(s) is/are correct?

(a) From above graph, it is evident that mole fraction (A) in vapour state is always higher than (B) irrespective of composition of solution.

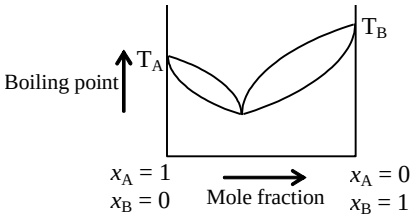
(b) Graph of boiling point vs mole fraction for this solution would be



(c) Graph of boiling point vs mole fraction for this solution would be:



(d) If (A) and (B) solution have shown positive deviation then their boiling point vs mole fraction graph would have been:



8. In which of the following reaction N_2 gas is produced.

(a) $NH_4Cl + NaNO_2 \xrightarrow{\Delta}$ (b) $CaOCl_2 + NH_3 \xrightarrow{\Delta}$

(c) $(NH_4)_2Cr_2O_7 \xrightarrow{\Delta}$ (d) $CaCN_2 + H_2O \xrightarrow{\Delta}$

SECTION – III
Reasoning Type

This section contains 4 reasoning type questions. Each question contains Statement-1 and Statement-2. Each question has 4 choices (a), (b), (c) and (d), out of which **ONLY ONE** is correct.

Directions: Read the following questions and choose

(A) **Statement-1** is True, **Statement-2** is True ; **Statement-2** is a correct explanation for **Statement-1**.

(B) **Statement-1** is True, **Statement-2** is True ; **Statement-2** is not a correct explanation for **Statement-1**.

(C) **Statement-1** is True, **Statement-2** is False.

(D) **Statement-1** is False, **Statement-2** is True.

9. **Statement-1:** Hydrate of cyclopropanone is readily isolated from its aqueous solution.

Statement-2: Aldehydes and ketones, undergoes reversible hydration in aqueous solution.

(a) A (b) B (c) C (d) D

10. **Statement-1:** Van't Hoff factor (i), for Mercury (I) nitrate on its complete dissociation would be three.

Statement-2: Mercurous ion exist as a dinuclear metal ion.

(a) A (b) B (c) C (d) D

11. **Statement-1:** When electrolysis of H_2O and D_2O mixture is done, H_2 is liberated much faster than D_2 and the remaining water thus becomes enriched in D_2O.

Statement-2: Natural abundance of ^1_1H is nearly 99.9 times the ^2_1D.

(a) A (b) B (c) C (d) D

12. **Statement-1:** Barium salts are toxic to us, still examination of digestive tract by doctors are done by giving a 'cocktail' containing $BaSO_4$.

Statement-2: $BaSO_4$ is opaque to X-ray. So, digestive tract is easily examined by X-ray.

(a) A (b) B (c) C (d) D

SECTION–IV
Comprehension Type

This section contains 2 groups of questions. Each group has 3 multiple choice questions based on a paragraph. Each question has 4 choices: (a), (b), (c) and (d) for its answer, out of which **ONLY ONE** is correct.

Passage-I

0.369 g of a bromoderivative of a saturation hydrocarbon (A) when vaporized occupied 67.2 mL at STP. (A) on reaction with aqeuous NaOH gives compound (B). (B) when passed over alumina at 250°C gives a neutral compound (C) while at 350°C it gives hydrocarbon (D). (D) when heated with HBr gives an isomer of (A), which on treatment with alcoholic KCN followed by acidic hydrolysis gives (E). Treatment of (E) with one equivalent of Br_2/Red P (HVZ reaction) gives α-bromo acid which is optically inactive.

13. What is the molecular formula of (A)?

(a) C_3H_7Br (b) C_4H_9Br

(c) C_2H_5Br (d) $C_5H_{11}Br$

14. The compound (E) is

(a) $CH_3CH_2CH_2COOH$ (b) $(CH_3)_2CHCH_2CH_2COOH$

(c) CH_3CH_2COOH (d) $(CH_3)_2CHCOOH$

15. Compound (C) on reaction with one equivalent of HBr gives

(a) (A) only (b) Both (A) and (D)

(c) Both (B) and (D) (d) Both (A) and (B)

Passage-II

Some of the important applications of gases are dissussed in the following paragraph.

1. Most automobiles are now equipped with air bags to protect the driver and passangers in the event of a head-on or side crash. Such bags are inflated with nitrogen gas, which is generated by the explosive decomposition of sodium azide.

   $2NaN_3(s) \longrightarrow 2Na(s) + 3N_2(g)$

   The air bag is fully inflated in about 0.05 second. This is important because the typical automobile collision lasts about 0.125 seconds.
2. A hot air ballon rises because the heated air has a lower density than surroudning air, which result in upward movement of balloon. This form of transportation is now a days is used for meterological research. Density of gas is found as $d = \frac{m}{V} = \frac{PM}{RT}$.

   The siginificance of gas density has been revealed in several tragic events. Once in African country of Cameroon in 1984 when take Nyos expelled a huge bubble of CO_2 into the atomosphere. Because CO_2 is denser than air, the CO_2 cloud hugged the ground, killing 1700 people nearby.

16. You are asked to design an air bag for a car. You know that the bag should be filled with gas at a pressure higher than atmospheric pressure, say 829 mm Hg, at a temperature of 22°C. The bag has a volume of 45.5 L. What quantity of sodium azide should be used to generate the required quantity of gas?

(a) 22.2 g (b) 44.4 g (c) 88.8 g (d) 66.6 g

17. Helium filled balloons are used to carry scientific instruments high into the atmosphere. Suppose a balloon is launched when the temperature is 22.5°C and the barometric pressure is 754 mm of Hg. If the balloon's volume is 4.19×10^3 L (and no He escapes from balloon), then what will the volume of balloon at the height of 20 km where pressure is 76 mm of Hg and temperature is –33°C?

(a) 3.38×10^6 L (b) 3.82×10^2 L (c) 3.12×10^4 L (d) 3.65×10^5 L

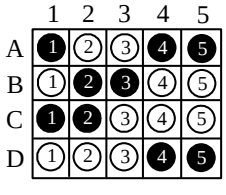
18. Which of the following interpretation(s) or observation(s) are correct for extinguishing fire by CO_2 gas?

(a) CO_2 is denser than air that is why it settles on top of air and smothers it.
(b) When CO_2 released from tank, it expands and cools significantly.
(c) White cloud seen is a condensed moisture from the air, when CO_2 is released.
(d) All are correct.

SECTION–V
Matrix–Match Type

This section contains 2 questions. Each question contains statements given in two columns, which have to be matched. The statements in **Column I** are labelled A, B, C and D, while the statements in **Column II** are labelled 1, 2, 3, 4 and 5. Any given statement in **Column I** can have correct matching with **ONE OR MORE** statement(s) in **Column II**. The appropriate bubbles corresponding to the answers to these questions have to be darkened as illustrated in the following example:

If the correct matches are A – 1, 4 and 5; B – 2 and 3; C – 1 and 2; and D – 4 and 5; then the correct darkening of bubbles will look like the following.



1.

Column I	Column II
(A) Sol of $Fe(OH)_3$ containing $FeCl_3$	**1.** Positive colloid.
(B) Sol of AgI containing KI	**2.** Negative colloid.
(C) Colloidal sol of gold metal	**3.** Formed by peptization.
(D) Sol of As_2S_3 containing H_2S	**4.** Formed by chemical method.
	5. Formed by Bredig arc method.

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2.

Column I	Column II
(A) ΔS (entropy change)	**1.** State function
(B) ΔH (enthalpy change)	**2.** Extensive property
(C) T (temperature)	**3.** $C_p dT$
(D) V (volume)	**4.** $\left(\frac{\partial H}{\partial S}\right)_P$
	5. Intensive property

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